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What is the A-GPS Tracker App?

A-GPS Tracker can manage your position and altitude using the A-GPS capabilities of your phone. When track recording is started it may trace and monitor your path, remaining active even when the phone goes to stand-by or when you are using the phone for other applications.

A-GPS Tracker is designed in particular for hikers that want to record a new path or want to follow an existing track and, even more important in my experience, if they want to avoid losing themselves during their return way.

Maps are made available free of charge by Google but only online. Google does not allow them to be used offline.

Another version of this App: AGPS-Tracker++ is using offline maps as made available by the Open Street Map project.

Tracker Features

The following **user geo-coordinates** are given in real time:

Elevation: is given in meters and is relative to Mean Sea Level .Note that most android programs are only giving the raw data provided by GPS ,i.e. height relative to a geocentric reference ellipsoid (WGS84).

Latitude and Longitude coordinates are given both in degrees and in UTM-WGS84 (used in some published paper maps).

A Track can be displayed on the map, recorded, stored or loaded to/from phone memory and shared with other people.

Points Of Interest(also called Waypoints) can be added to your track.

The Elevation vs Distance profile of a track can be shown on the screen.

Statistics of a loaded GPX track will include, among the others, the calculated net walk time, ascent and descent altitude.

A “Follow Path” monitor function can be enabled to get an alarm in case you are leaving a pre-loaded GPX path.

Take Photographs and see where the photos were taken on the map.

A-GPS Tracker is designed and normally works well on most Android devices, does not consume much battery power and you can leave it active for hours. With a limited number of phone types the App may not work and you will need to change your phone settings. If the path acquired shows long straight lines, this is the symptom that the device is not working correctly: the phone is set to power saving, pausing, or stopping the App when it is not displayed on the screen. In this case see Troubleshooting.

A-GPS Performance

GPS is used here as a generic term and refers to the capability to receive position data from signals received from satellites. Modern phones can obtain position data from more than one satellite system, i.e. not only the American GPS but also GLONASS, BeiDou and Galileo. A-GPS Tracker will receive data from all the supported and available satellite navigation systems; in addition A-GPS (Assisted GPS) also uses ground sources like cell tower data and WiFi, that may enhance quality and precision when in poor satellite signal conditions and in particular will help to reduce time to start fixing initial position.

GPS will be the source of the more accurate position: Latitude, Longitude and Altitude. Altitude provided by GPS is corrected (EGM96) to obtain altitude relative to mean sea level.

Note that for hikers Altitude it is of particular importance and can be given only by GPS. GPS may get poor position coordinates when GPS signals are weakened by meteorological conditions and may lead to significant errors in urban areas, when satellite signals may be reflected by buildings and therefore are received from a multi-path propagation.

You will get precise position and altitude when:

you are in an open sky place,
your phone antenna is not obstructed by metallic objects, buildings or mountains
and meteorological conditions are sufficiently good .

This initial GPS lock requires the view of a several satellites at the same time, A-GPS-Tracker will consider a good lock only if the elevation data will be available. Note also that elevation data is the less precise of the data provided by GPS, initially it may be quite inaccurate, but the continuous acquisition of new data will correct the initial reading. More in general consider that the elevation data will be strongly filtered and sharp changes in the elevation will be followed with delay. Note also that the performance of the GPS is variable and depends from many factors like the relative position of satellites. Before making any evaluation you should try more than once under the same settings.

GPX Files

A GPX File is a XML data format File, written according to the GPS Exchange Format for the interchange of GPS data (waypoints, routes, and tracks) between applications and Web services on the Internet.

The GPX file created by AGPS-Tracker has no references to photographs. The photographs are independent of the file and have embedded data related to the location and time of the shot.

Control Buttons Functionality

Compass indication. Normally the map is oriented north. By pressing this button, the map will be oriented in the direction indicated by the Compass. This is not present on phones that do not have the right sensors, i.e. magnetometer and gravity.

GPS is enabled and the signal is good! High precision geocoordinates will be sampled continuously. Recording of your path can run smoothly.

Push this button if you want your position (blue circle) in the center of the map.

Pushing this button you will be asked to select the type of map that you want to see. Default is Terrain.

The view of the map needs an internet connection. If you do not have it because you are in a wild environment, never mind it is actually not necessary to track your path.

By pressing this button you can take a picture. The relative symbol will be shown on the track.

This will open the **Track Recorder** window with the following buttons:

REC/STOP to start/stop and restart the recorder. Note that actual recording will start only if high precision geolocation coordinates and altitude will be available.

SAVE to save into a GPX file the path that you have been walking so far.

RESET to cancel your current path.



This button appears only when the Track Recorder has started to sample your position. With this you can add a Point Of Interest marker to your track.

This button appears only when an existing GPX file has been loaded. Push this button if you want the loaded track to be shown in the center of the map. When this is pushed, statistics relative to the loaded track are shown on a scrollable window.

This button will open another window showing the Elevation vs Distance profile. In this graphic drawing window you can tap a point on the graph and see also its position on the Map and the other way round.

Pushing this button you can see an icon moving on the map from the start to the end of the loaded track.

When this button is checked the phone will generate an alarm when the distance from the loaded track and the user position is larger than 30m. Alarm will generate a beep and a warning message.

Troubleshooting

The main problems happens when the screen is off (phone in stand-by) or showing another app, then you can have:

A-GPS Tracker application crashed, with no notifications.

Bad tracking with long straight lines in your path.

A-GPS Tracker is designed to remain active also when the phone goes in stand-by, like any other tracker application. But some phones try to kill this kind of apps that are showing a continuous power consumption, even if little. This behaviour can normally be disabled but unfortunately it is nested in the various settings of the phone and can be different for **manufacturers**, **models** and **Android versions**.

Some of these parameters have normally been set during A-GPS Tracker installation, but you shall check your phone to :

1. **Disable any Power saving for the phone**(Settings parameters or dedicated energy saving apps)
2. **Disable energy saving optimisations for A-GPS Tracker.**

The site <https://dontkillmyapp.com/> contains specific instructions for various phones and manufactures.

You can also find some instructions on https://agps-tracker.com/?page_id=2#phone_settings

Credits

Many thanks to the friends who have allowed the translation in different languages:

Alberto Ramotti
Jan Edvardsen
Renato Guimarães

Other Translations:



Español



Français



Italiano



Русский



Português do Brasil

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